Activity 0.1 – Everyday Task as an Algorithm

Title: From Daily Routine to Algorithm

Instructions to Students

1. Choose one simple daily activity (examples: making tea, preparing to go to class, booking a cab).
2. Write the steps in clear, numbered points.
3. Make sure each step is:
   - Unambiguous
   - In correct order
   - Does not skip important details

Template

1. Activity chosen: Making chocolates
2. Write the steps:
   1. Collect all the ingredients required for preparation
   2. Prepare the water bath. by swith on gas.
   3. Melting of the dark chocolate base in water bath till it become thi in consistnc
   4. Arrange the moulds.
   5. Smear the one layer of chocolate to the mould
   6. Allow it to freeze in freezer for 3min
3. Check your algorithm:
   - Can a friend follow it without asking questions? Yes ✓ / No
   - If "No", refine the steps here:
     -

7) Take that mould. Add another layer of Honey and allow it to freeze in freezer for 3-4min

8) Again take out that mould and add another layer of dark chocolate base. allow it to freeze for 3-4 min

9) Then take ~~away~~ out the moulds and collect the ready chocolates

10) Then wrap the chocolates with the chocolate wrappers

11) Then chocolate is ready to eat.

Activity 0.2 – From Steps to Pseudo-code

Title: Writing Pseudo-code with IF and REPEAT

Instructions

1. Take your steps from Activity 0.1.
2. Identify where decisions happen (IF/ELSE).
3. Identify where repetition happens (REPEAT / WHILE).
4. Rewrite your algorithm using the following keywords:
   - IF, ELSE, ELSE IF
   - REPEAT, WHILE, UNTIL

Template

1. Original activity: Making chocolate
2. Pseudo-code version:

text

START

IF chocolate is completely melted THEN

add it to the chocolate mould

ELSE

allow it to melt completely

ENDIF

REPEAT

Freezing layer wise

UNTIL Comple 3 layer is Set

END

3. Reflection:
   - Where did you use IF/ELSE? chocolate melting
   - Where did you use REPEAT/WHILE? Freezing of chocolate

Activity 1.1 – Run and Edit Given Code

Title: First Python Script – Edit and Run

Instructions

1. Your teacher will give you a small Python script (example below).
2. Do NOT write new code first; only change text or numbers.
3. Run the program and observe the output.

Example Script

python

```
name = "Student"
age = 18
print("Hello,", name)
print("You are", age, "years old.")
```

Tasks

1. Change name and age to your own values.
    - name: "Reshma"
    - age: 21
2. Run the program. What is the output?

text

Output:

"Hello Reshma"

"You are 21 years old"

3. Change the printed message to:
    - “Welcome to Python, <your name>”

Write your edited code here:

python

 # Edited code:

```
name = "Reshma"
Print ("Welcome to python," name )
```

out put:

"Welcome to python, Reshma"

4. Reflection:
    - What changed in the output and why?

The printed message changed in the out put. because we texted diffrnt message

Activity 1.2 – Predict-Then-Run

Title: Predict Program Output

Instructions

1. Look at each code snippet.
2. BEFORE running, write what you think the output will be.
3. Then run it and compare.

Snippet A

```python
x = 5
y = 2
z = x + y * 3
print(z)
```

- Predicted output: 21
- Actual output: 11

Snippet B

```python
a = 10
b = a - 3
a = a + 1
print(a, b)
```

- Predicted output: (11, 7)
- Actual output: ~~( )~~ (11, 7)

Reflection

- Where was your prediction wrong? Why?

I followed BODMAS rule Snippet A. because

Activity 2.1 – Calculator in Steps

Title: From Word Problem to Code

Problem 1
A student has marks in 3 subjects. Find total and average.

1. Write the steps in plain language:
   1. Write the marks in 3 Subjects (ie a,b,c)
   2. Take the sum of 3 marks, ie total = a+b+c
   3. To take average divide the total with total number of Subjects
   4. Average = $\frac{a+b+c}{3}$
2. Now convert to Python (fill in the blanks):

python

mark1 = 15 ⇒ a

mark2 = 18 ⇒ b

mark3 = 20 ⇒ c

total = 53 ⇒ a + b + c

average = 17.6 ⇒ $\frac{a+b+c}{3}$

print("Total =", total)

print("Average =", average)

3. Test with your own numbers and record output.
   a=15, b = 18 c = 20 total = 19 + 18 + 20 = 53
   ~~3~~ Average = $\frac{53}{3}$ = 17.6

Activity 2.2 – Fix the Errors

Title: Debugging Simple Python

Instructions
Each snippet has an error.

1. Write what you think is wrong.
2. Correct the code.

Snippet 1

python

print("Hello World)

- Error: Double inverted comma
- Corrected code: print ("Hello World")

python

undefined

Snippet 2

python

a = 10

b = "5"

c = a + b

print(c)

- Error: b is not astring , double inverted comma is not used
- Corrected code: a = 10
  b = 5
  print (c)

python

undefined

Reflection:

- What did you learn about syntax vs logic errors?

Syntax error does not makes the program tobe executed logical errors are simple and it will affect the output not

Activity 3.1 – Rule-based Classification

Title: Grades with IF/ELSE

Rule

- If mark ≥ 90 : Grade A
- Else if mark ≥ 75 : Grade B
- Else : Grade C

Tasks

1. Draw the decision in simple text:
   - Condition 1: if marks is more than or equal to 90 → grade A
   - Condition 2: if mark is more than or equal to 75 → grade B
     or else grade C
2. Now write Python code:

python

mark = int(input("Enter mark: "))

if mark >= 90:

print("Grade A")

elif mark >= 75:

print("Grade B")

else:

print("Grade C")

3. Test with at least 3 marks and note outputs.

if mark is 95
Out put → Grade A

Mark = 81
Out put → Grade B

Mark = 39
Out put → Grade C

Activity 3.2 – Mini Problem Bank

Title: Structured Problem Solving with IF/ELSE

For each question, fill the template.

Template

1. Restate the problem in one sentence: Sowing Seeds based on rainfall prediction (RF)
2. Conditions needed (bullets): (RF ≥500mm)
   - if rain comes sow the seeds
   - ~~if~~ else no sowing of seeds
3. Python code:

python

 # Your solution

4. Test cases and results:
   - Input: 650 Output: Sow the seeds
   - Input: 250 Output: No Sowing of Seeds

Use this template for:

- Odd or even number.
- Ticket discount if age < 12 or > 60.
- Check if a year is a leap year (simple rule).

---

1) Odd oreven number

```
number = int (input ("Enter number"))
if number,%2==0 :
   print ("Even")
else :
   print ("odd")
```

2)

```
age = int (input ("Enter age"))
if age <12 ~~or >60~~ :
print (" Discount in ticket")
elif age >60 :
print (" dis count in tick
else:
print (" No discount int
```

3) Check if a year is a leap year

```
Year = int (input (" Enter year"))
if year % 4 ==0 :
 print (" Year is leap year")
else :
 print (" Not leap year")
```

Activity 4.1 – Sum

Title: Sum from 1 to

Steps

1. In plain lan

Activity 4.1 – Sum with Loops

Title: Sum from 1 to N

Steps

1. In plain language:
   - What is "sum from 1 to N"? 1+2+3+4 .... +N
2. Write the algorithm:

1. First write / compute (N+1)
2. Then multiply N with (N+1).
3. Then divide the product by 2.
4. Then the Quotient value is the sum of number

3. Now code it:

python

```
N = int(input("Enter N: "))
total = 0
for i in range(1 , N+1):
    total = total + i
print("Sum =", total)
```

4. Try N = 5, 10, 100 and check results.

1) For N = 5

$$Sum = \frac{5(6)}{2} = \frac{30}{2} = 15$$

2) For N = 10

$$Sum = \frac{10 \times 11}{2} = \frac{110}{2} = 55$$

3) For N = 100

$$Sum = \frac{100 \times 101}{2} = \frac{10100}{2} = 5050$$

Activity 4.2 – Star Patterns

Title: Loops for Patterns

Pattern (triangle)

```
*
**
***
****
```

1. Explain the pattern in words:

The pattern is the triangle.

2. Write code:

python

```
rows = int(input("Enter number of rows: "))
for i in range(1, rows + 1):
  print("*" * , ~~end=""~~) i)
```

3. Try different rows and observe.

```
row = int (input ("Enter number of rows:")
columns = int (input (" Enter number of columns:")
 for i in range (3)
 for j in range (3)
   print (" * ", end =" ")
   print ()
```

Activity 5.1 – Working with Lists

Title: Analyze Marks List

Given

python

marks = [56, 78, 90, 45, 88]

1. In words, write how to find:
   - Maximum mark: mox than any other mark in a list
   - Average mark: Total mark /5
   - Count of marks ≥ 60: which is mox than or equal to 60

2. Now write Python:

```
python

marks = [56, 78, 90, 45, 88]


# Max

max_mark = 90


# Average

total = 0

for m in marks:

    total = 56+78+90+45+88

average = total /5


# Count >= 60

count = 0

for m in marks:

    if Count 7/60:

        count = 3


print("Max:", max_mark)

print("Average:", average)

print("Count >= 60:", count)
```

Activity 5.2 – Choosing an Approach

Title: Which Method is Better?

Problem

Given a list of 1000 numbers, find how many are positive.

1. Approach A (loop) – describe in words:

   for loop

2. Approach B (using built-in functions or list comprehension) – describe:

   if else function if no. is >0 is +ve or else -ve

3. Which is easier to understand for you now? Why?

   built in function is easy to understand

4. Implement one approach in Python:

python

numbers = [10, -3, 5, 0, 8, -2] *# example*

 # Your code

```
number = int (input ("Enter number "))
 if number >0 :
  print (" number is positive ")
 else:
   print ("number is negative")
```

Activity 8.1 – Creating a Function

Title: Wrap Logic into a Function

Task

Turn average-of-marks logic into a function.

1. In words: what does the function do?

2. Write code:

python

def average_marks(marks_list):

total = 0

for m in marks_list:

total = total + marks (m)

avg = total/ len mark list

return avg

data = [60, 70, 80]

print("Average =", average_marks(data))

3. Test with another list and note output.

data = [10,20,15]

print ("Average = ", Average_marks (data))

Out put :

Average = 15

def average _marks (marks _list):

total =0

for m in marks _list:

total = total +m

if len (marks_list) >0:

avg = total / len(marks_list)

return avg

else:

return =0

Activity 6.2 – Problem-Solving Template

Title: Standard Template for Any Problem

Use this for any new problem.

1. Problem understanding
   - What is given (inputs)?

Age is given

   - What is required (outputs)?

wether discount is applicable or not

   - Any constraints (limits, ranges)?

b/w 12-60 years of age no discount applicable

2. Analysis
   - Write steps in bullets (algorithm):
     - if age is <12 50% discount
     - elif age is >60 30% discount
     - else no dis count
3. Design
   - Will you use loops? Which? No
   - Will you use conditions (if/else)? Yes
   - Will you use functions? Which ones? No
   - What data structures (list, etc.)? No
4. Code (Python)

python

 # Your implementation

5. Test
   - Test case 1: input 11 → output Discount applicable
   - Test case 2: input 53 → output No discount
   - Edge case: input 63 → output Dis count applicable

Activity 7.1 – Capstone Problem

Title: Simple Fee Calculator

Problem

A class charges a base fee.

- If student is below 12 years, 50% discount.
- If above 60, 30% discount.
- Otherwise, no discount.

Use the full template from Activity 6.2 here.

(Attach a blank copy of the template and ask students to fill all five sections.)

```
age = int(input("Enter age"))
if age <12 :
  print("50% discount")
elif age >60:
  print("30% discount")
else :
print("No discount")
```

Activity 7.2 – Reflection Sheet

Title: How I Solve a New Problem

1. When I see a new problem, my first step is:

Problem understanding → Check input, outputs & limits/ranges if in case present

2. Before coding, I make sure I know:

- Inputs: ______________________
- Outputs: ______________________

3. The tools I think about (loops, if, functions) are:

loops - for, while, if else data structure list, set, dictionary, tupel. etc.

4. One thing I will do differently from now on:

Cleary understand the program problem to be solved